CYBERMIND AI

UI guide: every screen, button and word explained

This guide explains what each part of the console and the demo app means, in plain words, followed by the full project flow. Where something is called a name that sounds technical, the meaning is given first.

The project in 30 seconds

Companies have a login page and, after login, a session (the period a person stays signed in). Attackers try to break in at the login (stolen passwords, bots) or steal an existing session. CYBERMIND AI watches both. It scores how risky each login or session looks, then **allows** it, asks for **extra proof**, **contains** it in a fake environment, or **blocks** it, and it shows the reasons to an admin on this console.

Words you will see everywhere

Session	One signed-in visit of one user. It has an ID like 607cc605 (only the first 8 characters are shown), an IP address and a device.
IP address	The internet address a request comes from. A change of IP in the middle of a session can mean someone else took over.
Device / User-Agent	The browser and operating system text a device sends, for example 'Chrome 155 on Windows'.
Behaviour signals	How the person types and uses the mouse: typing speed, gaps between keys, clicks, mouse movement. A bot or another person behaves differently from the real owner.
Rule	A fixed check with points, for example 'too many failed logins = 3 points'. Rules are easy to explain.
Baseline	What is normal for one particular user, learned from their own earlier behaviour. 'baseline_deviation' means they are far from their own normal.
ML / Isolation Forest	A machine learning model that spots behaviour that is statistically unusual overall. It adds 2.5 points when it says 'anomaly'. It is one signal, not the judge.
Score and thresholds	All points are added up. 3 or more = SUSPICIOUS, 6 or more = LOCKDOWN (the default settings). The number is for audit; people see the reasons.
SAFE / SUSPICIOUS / LOCKDOWN	The three status colours: green (normal), amber (watch closely, ask for proof), red (frozen).
Step-up	Asking the user to prove it is really them again, for example re-entering their password, before continuing.
Lockdown	The session is frozen. Everything protected is blocked until an admin unlocks it.
Client fingerprint	A short code for the attacker's software (browser or script, and the order of its request headers), not the IP. An attacker can change IPs thousands of times but the fingerprint stays the same, so it links many attempts together.
Campaign	A group of login attempts that share one fingerprint and look like an attack (stuffing, spraying). 'watching' = suspicious, 'blocked' = stopped.
Credential stuffing / spraying	Stuffing: trying leaked username and password pairs on many sites. Spraying: trying one common password against many usernames.
Decoy account	A fake username no real person uses. Anyone who tries it is an attacker, so it is an instant alarm.

Breached password	A password found in public leaks. We check it using a hash (a one-way scrambled form), so the real password is never stored or sent.
MITRE ATT&CK	A public catalogue of attacker techniques with IDs like T1090 (proxy and IP hopping) and T1499 (flooding a service).
SIEM	A company's central security log system (Splunk, Sentinel, Wazuh). The '.jsonl' export is in a format those can read.
RAG	Retrieval-Augmented Generation: the assistant first finds the relevant parts of your uploaded PDFs, then writes the answer only from them.
CVE / NVD	CVE is the public ID of a known software vulnerability (CVE-2024-1234). NVD is the US government database we check it against.

What does 'sandbox' mean? (the question that got asked)

The word appears in two different places. Say which one you mean.

1. Shadow Mode sandbox	A safe, fake copy of the app that we show to a session we are not sure about. The person sees normal balances and 'successful' transfers, but **no real data changes and no real money moves**, and everything they try is recorded. We use it so a real customer is not locked out by mistake, and an attacker does not learn they were caught. Think of a cage that looks like a room.
2. 'Spawn target sandbox' button	In the Attack Simulator. It only creates a **throw-away test session** (named attacker_target_123) so you have something to attack in the demo. It is not a real user, and nothing about it is real.
Also: the Attack Lab	The Realistic Attack Lab and the stress test run in a temporary in-memory database that disappears after the run, so they never touch real data.

SCREEN 1

The left sidebar and the Overview

Left sidebar (always visible)

CYBERMIND AI / SECURITY CONSOLE	The logo and the name of the admin console.
STATUS box	The overall health in one word: SYSTEM SECURE (green), ELEVATED WATCH (amber) or LOCKDOWN ACTIVE (red). It follows the real sessions on screen.
MONITOR / RESPOND / VALIDATE / SETUP / ASSIST	Group labels. They follow an operator's day: watch, then act, then test the defence, then set up, then ask for help.
Overview, Detections, Email Analysis (1 to 3)	Watching and investigating.
Active Sessions (4)	Containing: unlock, verify, review shadow cases.
Attack Simulator (5)	Testing the defence safely.
Integrations (6)	How a company connects its app to CYBERMIND.
Analyst Assistant (7)	The AI chat and report tool.
Small number box (1 to 7)	A keyboard shortcut: press the number to open that screen.

Open customer demo app	Opens VaultApp, the demo bank, in a new tab. That is what a customer sees.
Scenic flight OFF / ON	OFF: screens open instantly. ON: the camera flies through the 3D network to each screen first. Just for looks.
Quick actions Ctrl K	A search box for commands: open any screen, toggle scenic flight, open VaultApp, log out.
Log out	Ends the admin session.
Left and right arrows, 'Esc', 'Back to network'	Move between screens, and go back to the 3D overview.

Overview screen

REAL-TIME ENGINE ACTIVE + SYSTEM SECURE	A banner saying the engine is running and the current overall state with a one-line explanation.
ENGINE / FEED / RESPONSE chips	ENGINE: it uses an Isolation Forest plus 6 rules. FEED: updates arrive live over a WebSocket (no refreshing). RESPONSE: auto-lockdown is on.
Live session constellation	A 3D picture where each dot is a monitored session. Teal = normal, red = locked. Drag to rotate. It is the same data as Active Sessions.
Threat index (gauge, 0 to 100)	A quick feel for danger right now. Recent events add to it (a lockdown adds a lot, suspicious a little, safe almost nothing) and each fades within a few minutes. If any session is locked it starts from 38. It is a visual indicator, not a security decision.
Active sessions	How many sessions are being monitored now.
Events / min	How many telemetry events arrived in the last minute (the small line is the trend).
Suspicious events	How many events the engine flagged amber.
Phishing emails	Emails analysed so far, and how many were analysed in total.
Lockdowns triggered	Sessions frozen automatically.
Event pulse (chart)	The last two minutes of events as bars, coloured safe / suspicious / lockdown.
Live activity feed	A running list of the latest events with their status. A red banner also pops up when something serious happens.
Demo / Red team controls	OPEN ATTACK SIMULATOR jumps to the simulator. START NORMAL TRAFFIC SIMULATION sends fake normal-user behaviour so the screens have data. CREATE DEMO SESSION adds one test session.
Hardening status	A checklist of security settings: strong JWT secret, strong admin password, CORS limited, client IP header not trusted, live feed needs admin token, telemetry needs a signed token, login rate-limited. A tick is good, a warning means it needs fixing before real use.
Detection model status + 'Learn from real traffic'	Says what data the ML model was trained on. After 200 real safe events have been recorded, the button retrains the model on them (it is greyed out before that).
Lockdown containment	A list of frozen sessions with an unlock button. 'No sessions are frozen' when it is empty.

SCREEN 2

Detections: the audit trail

Four tabs at the top: **EVENTS**, **CAMPAIGNS & DECOYS**, **POISON TRACES** and **LIVE TAIL**.

EVENTS tab: every decision the engine made

Four number cards	Total events, how many were SAFE, SUSPICIOUS, LOCKDOWN, with the share of the events currently loaded and how many were ML anomalies.
Filter buttons ALL / SAFE / SUSPICIOUS / SHADOW / LOCKDOWN	Show only events with that status.
Event type dropdown	All types, or only 'behavior' (normal telemetry), 'step_up_cleared' (admin said the extra check passed) and so on.
ML anomalies only	Show only events where the Isolation Forest said unusual.
Search box	Search by session, IP, event, flag or action.
CSV and SIEM (.jsonl) buttons	Download the log. CSV opens in Excel. SIEM is for a company security system.
Table columns	TIME (and how long ago), SESSION (first 8 characters), EVENT type, SOURCE (IP and device), FLAGS, VERDICT and ACTION.
FLAGS	Which rules fired, for example baseline_deviation, ip_or_device_change, request_burst. 'ml_anomaly' means the model flagged it.
VERDICT	The status colour. Under it, 'step-up' means the user was asked for proof, 'block' means it was frozen.
ACTION	What the system actually did, for example step_up_required or cleared_by_admin.
Arrow at the row end	Click the row to open details: full session ID, exact time, the full user-agent, the action, the list of reasons with their points ('+3 Burst of API requests...'), and a RAW EVENT section with the raw record.
Load more	Loads the next 100 older events.

CAMPAIGNS & DECOYS tab: attacks on the login

CyberMind Network card	Shows if this install is connected to the sharing hub (OFF by default), how many warnings it received and shared, and the last sync or error. Received hints add only a few points and never block alone.
Attack campaigns table	One row per detected campaign: its ID, status (watching, blocked, released), scale (attempts, IPs, usernames), signals (failure rate, breached passwords, decoy hits) and the reasons with points. A Release button appears on blocked ones in case it was a false alarm.
Decoy accounts	The fake usernames. 'untouched' means nobody tried them. Add decoy creates one, the cross removes one. A touch is an instant alarm and blocks that client.
Recent login attempts	TIME, USERNAME, RESULT (OK or FAILED), IP, FINGERPRINT (the short code), FLAGS (for example 'breached pw') and DECISION (allow, step-up or deny).

POISON TRACES tab: from phishing email to the login

What it is	A list of fake credentials planted from phishing emails. You enter them yourself on a fake page; we never send them anywhere. If one is ever tried at our login, the attacker's whole campaign is blocked.

Columns	CREDENTIAL (the fake username), STATUS (planted or used), FROM EMAIL (which analysed email), PLANTED (when), PHISHING TO USE (how long the attacker took), CAMPAIGN (the blocked campaign).
Retire	Switches that credential off. The timeline then reads: email analysed, planted, used, campaign blocked.

LIVE TAIL

A switch that refreshes the event table every 4 seconds so new events appear by themselves.

SCREENS 3 AND 4

Email Analysis and Active Sessions

Email Analysis

Analyse an email (intro)	Reads the headers, links and attachments of an email and explains why it is or is not phishing. Links are never opened and attachments never run. Only the result is stored, not the email.
AI analyst (Groq) switch + model dropdown	Optional second opinion from an AI model. 20b is faster, 120b is stronger. When ON, the email text goes to Groq (attachments never do). When OFF or not configured, everything stays on this server.
UPLOAD .EML / PASTE SOURCE	Two ways to give the email: a downloaded .eml file, or the full message source pasted as text. The grey text explains how to get it from Gmail or Outlook.
ANALYSE FILE / ANALYSE TEXT	Starts the analysis.
Verdict banner and RISK SCORE	PHISHING, SUSPICIOUS or LIKELY SAFE with the score, and the two score lines (suspicious and phishing thresholds).
AI panel	The AI analyst's opinion, shown separately. It never changes the rule-based verdict.
Plant credential	Appears for suspicious or phishing emails. Creates a unique fake username and password tied to this email. The password is shown only once (copy it now); only a hash is stored. See Poison Traces above.
Download PDF report / Markdown / Delete	Save the analysis as a report or remove it.
The message	From, Reply-To, Subject, Date and a SHA-256 fingerprint of the file.
Why this verdict	Each warning sign with its points and evidence, for example a lookalike domain, a mismatch between the shown link and the real link, or a free mailbox pretending to be a company. Green ticks list good signs.
Links	SHOWN AS (the visible text) versus REALLY GOES TO (the real address). A mismatch is a classic phishing sign.
Attachments	Names and hashes only; they are never opened.
Indicators to block	Copyable lists of domains, IPs, sender addresses and file hashes that a company can put in its mail filter or firewall.
MITRE ATT&CK and What to do	Which attacker techniques the email uses, and a numbered list of recommended steps.
Recent analyses	A table of past results; click a row to reopen it.

Active Sessions

SESSIONS | SHADOW QUEUE tabs	Sessions: all monitored sessions. Shadow queue: sessions we are not sure about, being held in the sandbox for review (a number shows how many).
Unlock all (n)	Appears only when sessions are locked. Unlocks every frozen session at once.
Five number cards	MONITORED, ACTIVE (normal), SUSPICIOUS (under closer watch), IN SHADOW (contained, awaiting review), LOCKDOWN (frozen, admin unlock needed).
Search + filter buttons + table/card icons	Find a session by user, ID, IP or device; filter by status; switch between a table and cards.
Table columns	SESSION (avatar letter, name, short ID), STATUS badge, IP ADDRESS, DEVICE.
Unlock button (red rows)	Admin unlocks a frozen session.
Verified button (amber rows)	Admin confirms the extra check succeeded, so the session returns to normal.
Detail panel (click a row)	Opens on the right with the full session ID (Copy button), last IP, device and the full user-agent text, plus an unlock or verified button.

Shadow queue (inside Active Sessions)

What is shown	Four cards: IN SHADOW NOW, FALSE POSITIVES (real customers we released), CONFIRMED ATTACKERS (frozen after review), EXPIRED (nobody reviewed in 30 minutes, so it went back to step-up).
Case card	The customer name, session, IP and a countdown timer (turns urgent near zero). Under it: the reasons with points, and 'WHAT THEY TRIED', for example '1 transfer totalling $4,000 to Offshore Account'.
Real customer: release	Says this is the real owner. They get back to normal and see a notice listing which transfers did NOT run. The new device is remembered so it is not shadowed again. Counts as a false positive.
Attacker: freeze	Confirms an attacker. The session is locked using the normal lockdown path.
Intent log	Every action in the sandbox in order: logins, balance views, transfers with recipient and amount.
Resolved cases	History of released, confirmed and expired cases and who resolved them.

SCREEN 5

Attack Simulator

A red-team toolbox: you play the attacker so you can watch the defence react. A yellow line warns that it is an authorised simulation using sandbox (test) sessions only. Two tabs: **MANUAL TECHNIQUES** and **REALISTIC ATTACK LAB**.

MANUAL TECHNIQUES tab

Target session dropdown	Pick which session to attack. Each entry shows the name, status and IP.
Target status badge	The target's current status (SAFE, SUSPICIOUS, LOCKDOWN).
+ Spawn target sandbox	Creates a brand new throw-away test session to attack. Not a real user.

Test protected action	Tries a sensitive action (like a money transfer) on the target to see whether the defence lets it through.
Risk gauge (half circle)	An estimate of the target's score out of 6, the lockdown threshold. The arc fills as the score rises: halfway = SUSPICIOUS ZONE, full = LOCKDOWN THRESHOLD REACHED.
Engagement progress (4 steps)	Recon (target chosen), Evasion (IP change done), Impact (request flood done), Lockdown (defence triggered). It marks the attacker's progress like a kill chain.
Technique cards T1090 and T1499	The two attacks. T1090 = switch IP or device in the middle of a session. T1499 = flood the API with requests.
T1090 fields and presets	New source IP and new device text. Presets fill in a Tor exit node or a datacenter proxy. The big button sends the traffic with that fake identity.
T1499 fields and presets	Number of requests, the endpoint, the source IP and the device. Presets: data-exfil flood and search flood. The big button fires the burst.
Progress bar and counters	REQUESTS done out of total, RATE per second, STATUS (ready, in flight, complete).
Terminal on the right	A command-line style log of everything sent and the system's replies. ALL/INFO/WARN/ERROR filters, an event count, COPY and CLEAR. Click 'payload' on a line to see the raw reply. A shadow notice appears here when the target gets contained.
Admin unlock target	Shows when the target is locked so you can keep testing.

REALISTIC ATTACK LAB tab

What it does	Replays whole attacks together with hundreds of normal users through the real detection code on a fast fake clock, using a temporary database. It reports honestly what was caught and what was not.
Scenarios list	Normal users only (control), naive burst, rotating IPs, low and slow, password spraying, human-like bot, stealth diluted, decoy touch, session thief with different rhythm, session thief copying the owner, session thief on a new device (shadow mode), real owner on a new laptop (shadow false positive). Each shows what we EXPECT; two are expected to be missed to show the limits.
Scale / Run	How big to make the test, and the run button. Running takes seconds to a minute.
Number cards	ATTACKS CAUGHT, ACCOUNTS COMPROMISED (versus how many with no defence), NORMAL USERS DENIED, SHADOW FALSE POSITIVES, EXTRA-PROOF PROMPTS (friction for real users).
Results table	Per scenario: the verdict (caught, caught late, missed, contained, false positive, clean), at which attempt the defence reacted, what share was stopped, accounts compromised and how many normal users were bothered.

Adaptive stress test (bottom card of the lab)

What it asks	'How much must an attacker change before one slips past?' It keeps adjusting six attacker settings (tries, IPs, duration, fingerprint style, breached passwords) with a simple genetic search.
Run stress test	Starts it (about a minute, with a percentage).
Result cards	VERDICT (held or evaded), FIRST SLIP AT (which variation got through first), COST TO THE ATTACKER (how many more attempts per stolen account than with no defence) and NORMAL USERS DENIED.

Best evader and Weakest spot	Describes the best attack found and which rule it stayed under, so you know what to improve.
Table	The top attacker settings found, how much got through, the peak score against the block line and which rules fired.

SCREENS 6 AND 7, AND THE DEMO APP

Integrations, Analyst Assistant and VaultApp

Integrations

How a company plugs CYBERMIND in	The customers never see CYBERMIND. The company's app reports two things: login attempts (from its server) and behaviour signals (from the browser). CYBERMIND answers with a decision the app enforces.
Four setup steps	Set an app key, add the login hook, add the session hook, create decoy accounts.
1 Login protection hook	Sample code the company's server runs after each login attempt. It sends the username, IP, result, browser info and a SHA-1 hash of the password (never the password). The reply is allow, step_up or deny.
2 Session protection hook	Sample steps: start a session, send behaviour every couple of seconds, and check before sensitive actions. The reply is SAFE, SUSPICIOUS or LOCKDOWN.
This instance	The API address, the mode (development or production) and how many decoy accounts are active.

Analyst Assistant

Indexed vectors counter	How many small text pieces from your uploaded PDFs are stored for search.
Intel uplink	Choose a PDF (an incident report, playbook or bulletin) and press Index. The reports are kept in a separate store that does not touch detection or logs.
Knowledge repository	The list of indexed files. 'Clear store' deletes them all.
Threat analysis terminal	Type a question, choose the AI model (Gemini or GPT-OSS via Groq), press Execute.
Briefing header + copy / export	A formatted advisory. Buttons copy it or export it as a Markdown file.
Execution pipeline line	The steps the agents followed, in order.
Threat level pill	The AI's own rating (low to critical). It may be missing if the report does not say.
Five report sections	01 Summary of the report, 02 Attacks performed, 03 Suggested actions (ordered by priority, marked 'from the report' or 'best practice, not in the report'), 04 CVEs and what defines them (checked against NVD, 'UNVERIFIED' if not found), 05 Agents involved (planner, retriever, analyst, reference checker, fact-checker).
Groundedness x/5 and hallucination flag	A fact-check score of how well the claims come from your documents. 'Strictly grounded' is good; 'hallucination flagged' means a claim was not found in the sources.
Verified source documents	Which uploaded files the answer used.

VaultApp: the customer demo bank

Sign in / Open account	A normal login and registration page. Behind it, every login attempt is sent to CYBERMIND.
Available balance and Everyday account	The fake bank account. No real money exists.
Protected by CYBERMIND	A badge showing the session is being monitored.
Recent activity	The list of past transactions.
New transfer: Recipient, Amount, Memo, Confirm transfer	A sensitive action. CYBERMIND checks the session before allowing it.
Small privacy note	Only typing rhythm and mouse patterns are measured; what you type is never read or sent.
Verify it's you (pop-up)	The step-up: asked to re-enter the password before continuing, when the session looks suspicious.
Security alert / Session frozen screen	The lockdown. A full-screen 'LOCKDOWN ENGAGED' message with the reason tags and an ADMIN UNLOCK button.
'Please redo these transfers' notice	Shown to a real customer after an admin releases them from shadow mode, listing the transfers that did not actually run.

THE WHOLE PROJECT

How it all flows, step by step

The main flow

1. A person signs in	They type a username and password in the app (VaultApp). The app asks CYBERMIND 'should I allow this?' before checking the password.
2. Login guard checks	Is the username a decoy or a planted poison credential? Is this client fingerprint part of a blocked campaign? If yes: deny or ask for extra proof. Otherwise: allow.
3. The attempt is recorded	Each attempt is stored with its fingerprint. The campaign engine counts attempts, failure rate, number of IPs, breached passwords and decoy hits. Too many warning points turns a fingerprint into a 'watching' and then 'blocked' campaign.
4. A session starts	After a good login the app starts a monitored session and gets a signed token. From now on the browser sends behaviour signals every couple of seconds.
5. The decision engine scores it	Rules (failed logins, IP or device change, request burst, typing speed, odd hour, too many resources), plus the per-user baseline, plus the ML model. All points are added: under 3 SAFE, 3 to 6 SUSPICIOUS, 6 or more LOCKDOWN. Every point comes with a written reason.
6. The response	SAFE: carry on. SUSPICIOUS: step-up (verify it's you). Identity changed but unsure: shadow mode (sandbox). LOCKDOWN: frozen, admin must unlock.
7. The console updates live	Everything is pushed over a WebSocket: the Overview numbers and sphere, the Detections log, the Active Sessions list and the red banner for serious events.
8. The admin decides	Unlock a frozen session, mark a step-up as verified, release or freeze a shadow case, release a wrongly blocked campaign, add decoys, retire a planted credential.
9. The system keeps learning	It builds each user's baseline from their own behaviour, and the model can be retrained after enough real safe events.

The side branches

Phishing branch	An analyst analyses a suspicious email (Email Analysis). If it is suspicious or phishing, they can plant a fake credential. If the attacker later tries it at the login, step 2 refuses it and the whole fingerprint campaign is blocked, with a timeline.
Shadow branch	In step 5, if the identity changed (new device and network, or a big departure from the user's normal) but the evidence is not conclusive, the session goes to the sandbox instead of being blocked. The admin reviews it within 30 minutes. If nobody does, it falls back to step-up.
Network branch	When this install blocks a campaign it can share a scrambled hash of the fingerprint with a hub. Other installs only treat it as a small hint after at least two installs agree. It can never block anyone alone.
Testing branch	The Attack Lab and stress test replay attacks against the same detection code in a temporary database, so you can see what it catches and what slips through.
Assistant branch	Separate from detection: upload PDFs, ask questions, get a report grounded in them with a fact-check.

A 3-minute demo order

1) Overview: say what the status and sphere mean. 2) Open VaultApp, sign in, show the balance. 3) Run the shadow demo script and show the case in Active Sessions, Shadow queue; press Release. 4) Email Analysis: analyse the sample phishing email, press Plant credential; run the poison demo; show Poison traces and the blocked campaign. 5) Attack Simulator: spawn a target, fire the IP change, then the API flood, and watch the gauge and the lockdown. 6) Realistic attack lab and the stress test.

Questions you may be asked

Why not just block every suspicious session?	Blocking tells an attacker they were caught and locks real customers out by mistake. Shadow mode keeps uncertain cases contained, watched and reversible.
Is the AI deciding who gets blocked?	No. Rules, the baseline and the ML model score the session. A language model is only used for second opinions, reports and answers, never for a verdict.
What if a real customer is flagged?	They are asked to verify, or released from shadow with a clear notice. The queue counts these false positives openly.
What does the ML model add?	It catches unusual behaviour that no single rule describes, but it is only one signal worth 2.5 points, so it cannot lock a session alone.
How is this different from malware tools like Intezer?	Those analyse files. This protects logins and sessions in real time and explains each decision.
What are the limits?	It is a research and demo project. It was tested on simulated traffic, and a slow stealthy attack or a perfect mimic of the owner can still pass. The stress test shows this openly.